NAME: ______________________________________ CLASS: ________ INDEX: _________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

Answers

BIOLOGY
9648/01

Paper 1 Multiple Choice

29th August 2016
1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

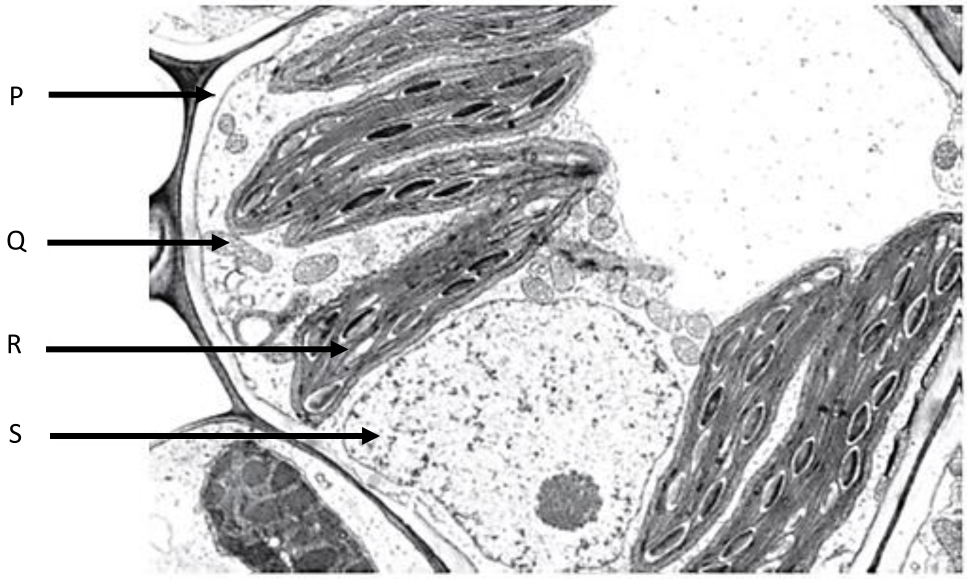
There are **forty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft 2B pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

1	B	2	C	3	C	4	A	5	C
6	C	7	D	8	A	9	C	10	B
11	B	12	C	13	D	14	B	15	C
16	C	17	D	18	D	19	C	20	A
21	D	22	C	23	C	24	B	25	B
26	D	27	B	28	B	29	D	30	A
31	B	32	D	33	D	34	C	35	C
36	D	37	D	38	D	39	A	40	A

1 The electron micrograph of a cell is shown below.



The electron micrograph of a cell is shown below.

Which of the following statements are true?

1 Structure P is found in all eukaryotic cells.
2 Organelle Q contains hydrolytic enzymes.
3 Organelle R contains starch.
4 Organelle S contains heterochromatin but not euchromatin.
5 Organelles Q, R and S contain RNA polymerase.

A 1 and 3 only

B 3 and 5 only

C 1, 3 and 5 only

D 2, 3 and 5 only

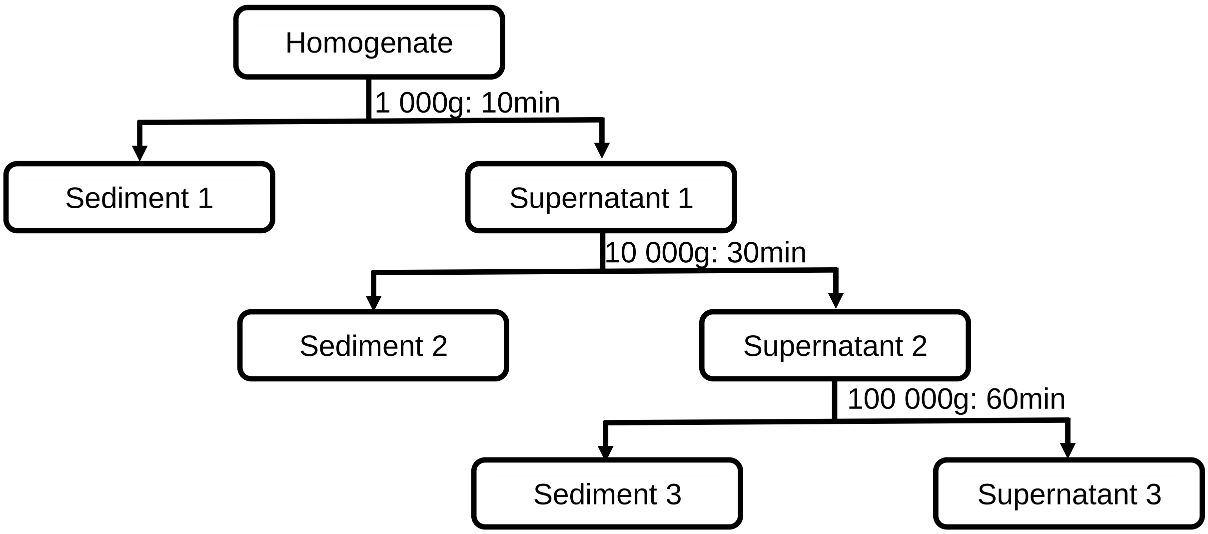
Ans: B

1.1] Cell structure and Organisation – EM

SC:	P- Cell Wall Q-Mitochondria R- Chloroplast	S- Nucleus
OR:	1 Structure P is found in all eukaryotic cells.	× Only Plant Cells
	2 Organelle Q contains hydrolytic enzymes.	× Q is not a lysosome
	3 Organelle R contains starch.	✓
	4 Organelle S contains heterochromatin but not euchromatin.	× Contains both
	5 Organelles Q, R and S contain RNA polymerase.	✓

[L1] (ACJC H1 PRELIM 2015 P1.Q1)

2 The figure below shows a centrifugation schematic of a rat liver cell.



Which of the following statements is incorrect?

A Sediment 1 contains organelles which are nucleic acid rich.

B Sediment 2 contains organelles with carbohydrate and nucleic acid.

C Supernatant 2 contains organelles that are the most dense amongst all other organelles.

D Supernatant 3 contains organelles that are involved in protein synthesis.

SC: Sed1: Nucleus Sed2: Mitochondria and Chloroplast Sed3: ER, Golgi

OR:

A Sediment 1 contains organelles which are nucleic acid rich — True

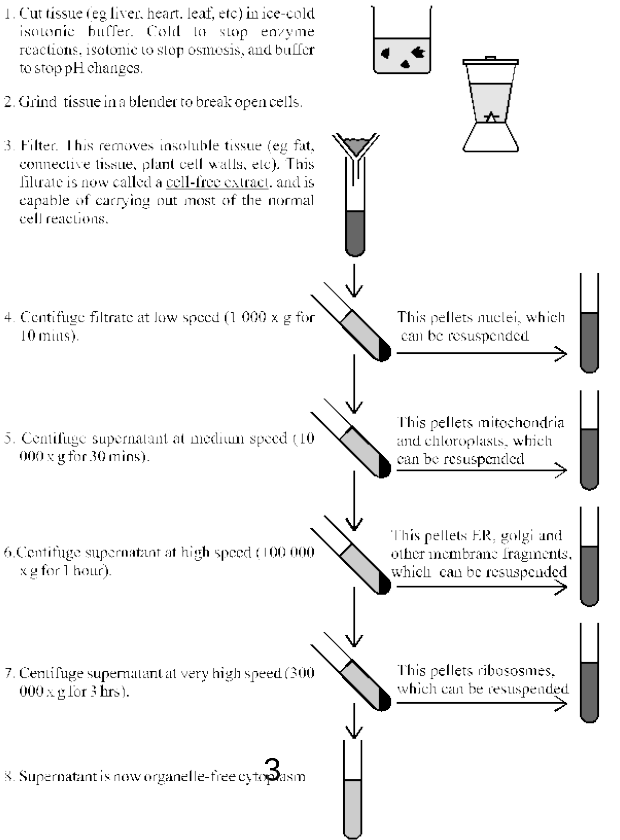
B Sediment 2 contains organelles which produce carbohydrate and nucleic acid — True
ATP is a nucleic acid

C Supernatant 2 contains organelles that are the most dense amongst all other organelles. — False
nuclei are in Sediment 1

D Supernatant 3 contains organelles that are involved in protein synthesis - True-
ribosomes

Ans: C

[L2] Novel



3 Which features of collagen result in it having high tensile strength?

1 covalent bonds form between adjacent molecules
2 each three-stranded molecule is held together by intramolecular hydrogen bonds
3 every third amino acid in the polypeptide is small
4 the primary structure is held together by peptide bonds

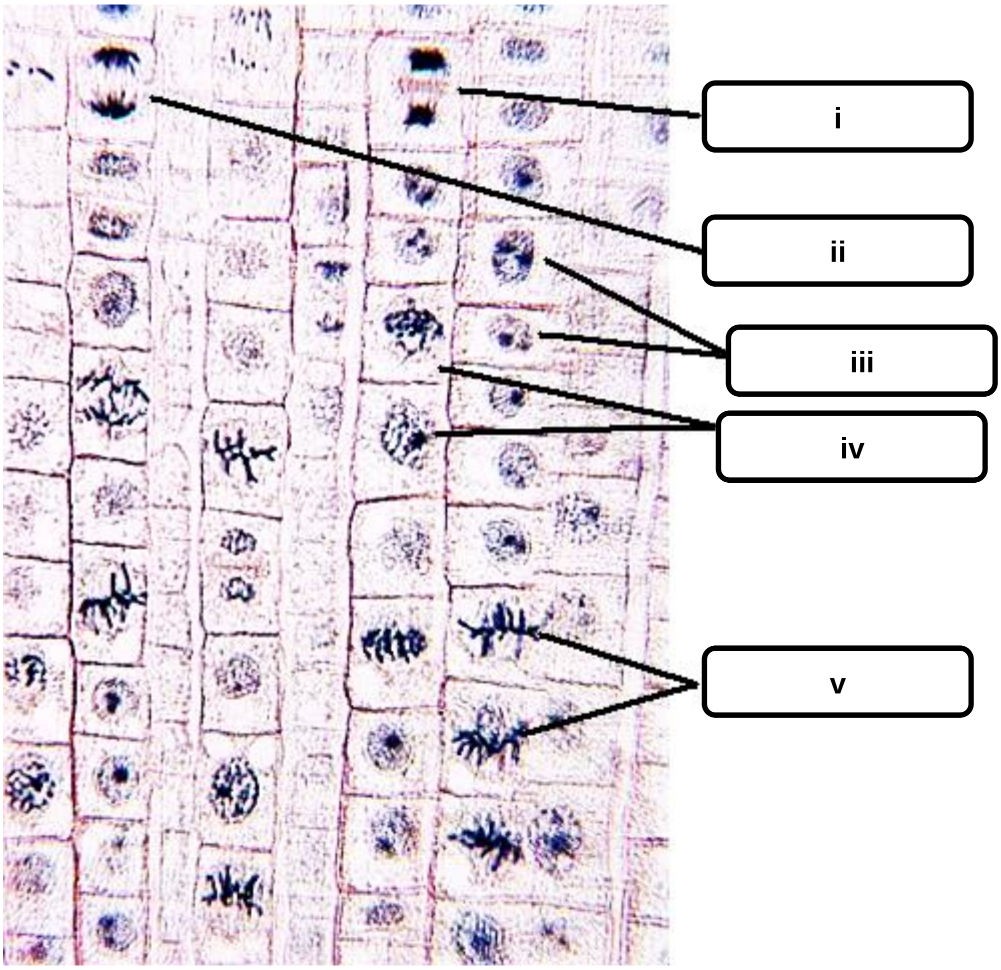
A 1 and 2
B 1, 2 and 3
C 1, 3 and 4
D All of the above

SC: features / collagen / high tensile strength
OR: Option 1 ✓ parallel strands bring cyctine groups together forming Disulphide cross bridges
Option 2 × No hydrogen bonds across molecules
(where present will more likely be intramolecular)
Option 3✓ X,Y,Glycine the third being small allows a tight coil in each fibril
Option 4✓ Primary structure is a polypeptide chain comprising of peptide bonds.

Ans: **C**

[L2] (2015 IJC Prelim P1Q5)

4 The diagram below shows the longitudinal section of a root tip.



Which of the following correctly outlines the sequence in the stages of cell division in the root tip

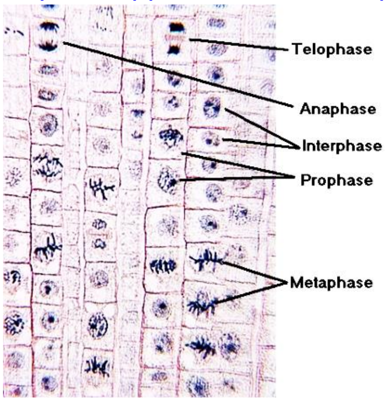
A iii > iv > v > ii > i

B iii > iv > i > v > ii

C iv > iii > v > ii > i

D iv > v > iii > i > ii

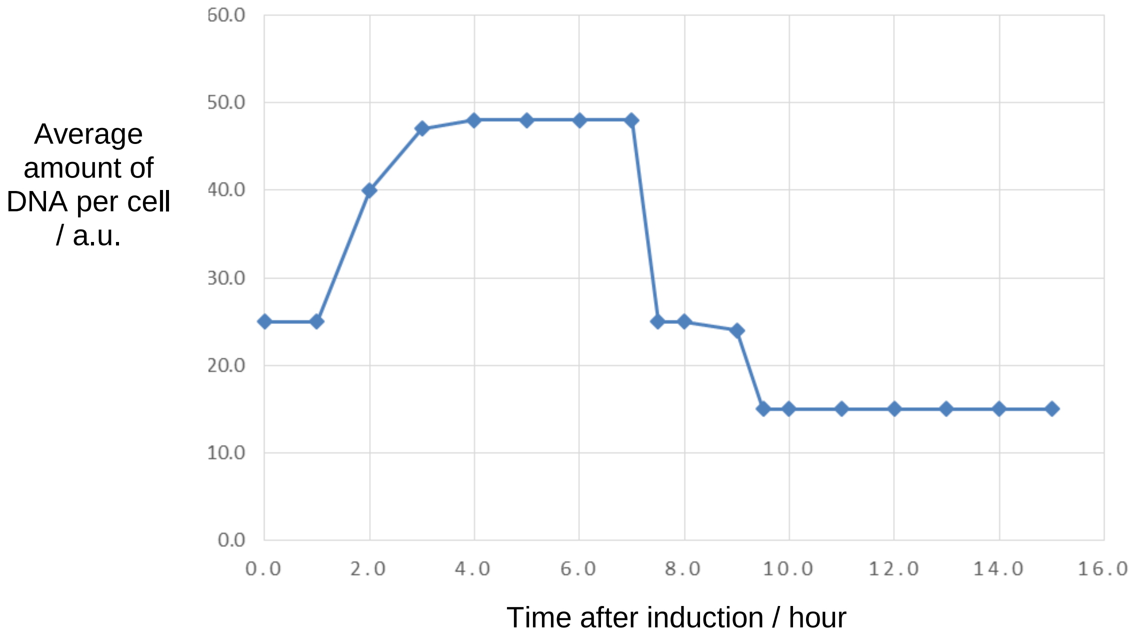
SC: correctly outlines seq in cell division
OR: Interphase (iii) >> Prophase (iv) >> Metaphase (v) >> Anaphase (ii) >> Telomerase (i)



Ans: **A**

[L2] Novel

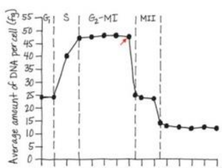
5 The figure below shows the average amount of DNA in a cell after induction.



Which of the following correctly accounts for the trends seen?

	Time Frame after induction / hours	Ploidy level at the end of timeframe	Stage in Cell growth
A	0.0 to 1.0	2n	G2
B	1.0 to 3.0	4n	S
C	3.0 to 8.0	n	G2-Meiosis I
D	8.0 to 9.0	2n	Meiosis II

SC: cell after induction / correctly accounts
OR: A × Ploidy 2n ✓ stage is G1 not G2
B × Ploidy 4n × stage S ✓ slow rise in DNA as replication increases DNA 2 fold
C✓ Ploidy n✓ since after cytokinesis time 7 hrs Stage G2-MI✓



D✗ Ploidy 2n✗ sice it is after cytokinesis time 7 hrs Stage MII. ✓

Ans: C
[L2] Novel

6 Which of the following is not required for transcription?

A Ribonucleoside triphosphates

B RNA polymerase

C RNA primer

D TATA box

SC: Need to know which factors play a role in transcription and which does not.
OR:
- ✓ A: Ribonucleoside triphosphates are the incoming monomers of transcription.
- ✓ B: RNA polymerase catalyses the process of transcription.
- ✗ C: RNA primer is not required. It is required only for DNA replication.
- ✓ D: TATA box is where RNA polymerase and general transcription factors will bind to during initiation of transcription.

Ans: C

[L1] Novel

7 A mutation had occurred on the template DNA strand which resulted in the polypeptide having the following sequence:

Met – Ser – Cys – Gly – Glu – Gln – His – Phe – Arg – Gly – Stop

The mRNA codon table is shown below.

First Letter	Second Letter U	C	A	G	Third Letter
U	phenylalanine	serine	tyrosine	cysteine	U
	phenylalanine	serine	tyrosine	cysteine	C
	leucine	serine	stop	stop	A
	leucine	serine	stop	tryptophan	G
C	leucine	proline	histidine	arginine	U
	leucine	proline	histidine	arginine	C
	leucine	proline	glutamine	arginine	A
	leucine	proline	glutamine	arginine	G
A	isoleucine	threonine	asparagine	serine	U
	isoleucine	threonine	asparagine	serine	C
	isoleucine	threonine	lysine	arginine	A
	methionine	threonine	lysine	arginine	G
G	valine	alanine	aspartate	glycine	U
	valine	alanine	aspartate	glycine	C
	valine	alanine	glutamate	glycine	A
	valine	alanine	glutamate	glycine	G

If the normal non-mutated template DNA strand has the following sequence,

3' – TAC – TCA – ACA – ACC – TCT – TGT – CGT – GAA – GGC – CCA – ACT – 5'

Identify the mutation(s) that had occurred.

A Single base pair substitution

B Deletion

C Addition

D Deletion and addition

SC:
Template: 3' –TAC–TCA–ACA–ACC–TCT–TGT–CGT–GAA–GGC–CCA--ACT–5'
mRNA: 5'- AUG--AGU-UGU-UGG-AGA-ACA-GCA-CUU-CCG-GGU -UGA-3'
Protein: Met – Ser – Cys -Trp-Arg-Thr-Ala-Leu-Pro-Gly-Stop
Mutated protein: Met– Ser– Cys– Gly– Glu– Gln– His– Phe– Arg– Gly– Stop

OR:
- A: If it is a single base pair substitution, there will only be one amino acid changing.
- B & C: If is is addition of deletion the frame will not be reinstated at the back.
- D: Most likely an addition occurred which caused a frameshift mutation. This is followed by a deletion which caused a resotring of the frame towards the end of the amino acid sequence.

Ans: D

[L3] Novel

8 The bacterium, Pneumococcus pneumoniae, forms two types of colonies whose cells are structurally different. Smooth (S) cells have thick outer capsules, but rough (R) cells lack this capsule. S cells cause the disease pneumonia.

In 1928, Frederick Griffith found that:

- when R cells were mixed with heat-killed S cells and the mixture injected into mice, some of the mice became infected and died.
- living S cells with capsules could be isolated from these dead mice.
- injection of heat-killed S cells alone or of living R cells alone did not cause disease in mice.

What can be concluded from these three observations to explain what happened when R cells were mixed with heat-killed S cells?

TRANSFORM

A A heritable genetic change occurred in the R cells.

B R and S cells conjugated when mixed.

C R cells were changed into S cells by transduction.

D R cells were transformed by DNA from heat-killed S cells.

SC: Process here is transformation.
OR:
- A: It is not mutation here and furthermore this does not explain what happened to the R cells.
- B: S cells were heat killed so it is not alive and conjugation requires sex pillus to form between two live bacterial cells.
- C: No virues were cited in the preamble and therefore transduction is possible.
- D: R cells underwent genetic recombination when it was transformed by a naked DNA that came from heat-killed S cells.

Ans: D

[L2] ('13 A-level/P1/Q13)

9 Which statements about bacterial conjugation are correct?

1 The F-plasmid is transferred to the recipient bacterial cell via the rolling circle mechanism.

2 An F plasmid carries genes controlling the process of conjugation.

3 Only one DNA strands of an F plasmid in the donor cell break at the origin of replication.

4 An F plasmid DNA strand enters the recipient cell beginning at its 5' end.

5 After transfer of F plasmid DNA, complementary strands of F plasmid DNA are synthesised in both donor and recipient cells.

6 Exonucleases cleave the donor DNA to create a nick.

A 1, 2, 3 and 5 only

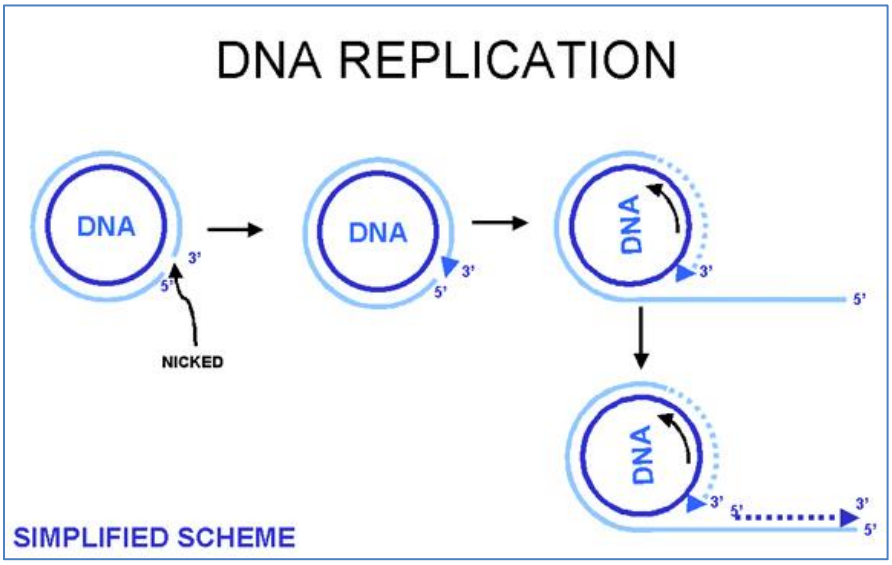
B 1, 2, 3, 5 and 6 only

C 1, 2, 3, 4 and 5 only

D All of the above

SC: Process here is conjugation. (Question was scaffolded in MYE & June holiday assignment)
OR:
- ✓1: Yes, the F-plasmid is transferred to recipient bacterial cell via rolling circle mechanism.

- ✓2: Yes, the F-plasmid carries the genes responsible for conjugation to take place.
- ✓3: Yes, only 1 strand break.
- ✓4: Yes, the F-plasmid enters the recipient cells from the 5' end onward.



- ✓5: Yes, the single-stranded DNA serves as a template for the second strand to be complementarily synthesized for the molecule to become double-stranded.
- ✓6: Yes, that is correct.
- ✗7: No. It is supposed to be endonucleases.
-

Ans: C

[L2] (Adapted from '14 A-level/P1/Q13)

10 Which of the following is not part of the Trp operon?

A Structural genes (*trp A* to *E*)

B *trp R*

C *trp* operator

D *trp* promoter

SC: *Trp* operon knowledge.
OR:
- *trp R* codes for the regulatory protein (Trp repressor) and it is not part of the Trp operon.

Ans: B

[L1] (novel)

11 Which statements about inducible and repressible systems are correct?

1. Repressible systems code for the synthesis of enzymes involved in anabolic pathways.
2. An inducible system is one where the operon is switched on under normal conditions.
3. An repressible system is one where the operon is switched off under normal conditions.
4. Inducible systems functions in catabolic pathways, digesting nutrients to simpler molecules.
5. An example of a inducible system is the Trp operon.

A All of the above

B 1 and 4 only

C 2, 3 and 4 only

D 1, 2, 3 and 4 only

SC: Inducible and repressible systems
OR:
- ✓1. Repressible genes code for the synthesis of enzymes involved in anabolic pathways.
- ✗2: An inducible system is one where the operon is switched off under normal conditions.
- ✗3: An repressible system is one where the operon is switched on under normal conditions.
- ✓4: True.
- ✗5: An example of a inducible system is the Lac operon.

Ans: B

[L2] (novel)

12 Viruses are considered obligate parasites because

A they reproduce using host cell DNA polymerase.

B they lack RNA-dependent RNA polymerases and ribosomes hence must depend on the host cell to carry out gene expression.

C they are unable to generate or store energy in the form of ATP and thus derive their energy for all metabolic functions from the host cell.

D they make use of the host cell's inorganic molecules such as amino acids, nucleotide and tRNA.

SC: Inducible and repressible systems
OR:
- ✗A. Reproduction does not occur on host cell ribosome.
- ✗B: Should be DNA-dependent RNA polymerases.
- ✓C: Correct.
- ✗D: Should be organic instead of inorganic.

Ans: C

[L1] (novel)

13 Which of the following proposed methods would be most viable in treating influenza?

A Introducing a ribosome inhibitor so that translation of viral proteins cannot take place.

B Introducing a ribonucleotide analog that would cause chain termination upon addition to an RNA polymer.

C Inhibit the attachment and thereby entry of the virus into its host cell by developing inhibitors that bind to the sialic acid on host cells.

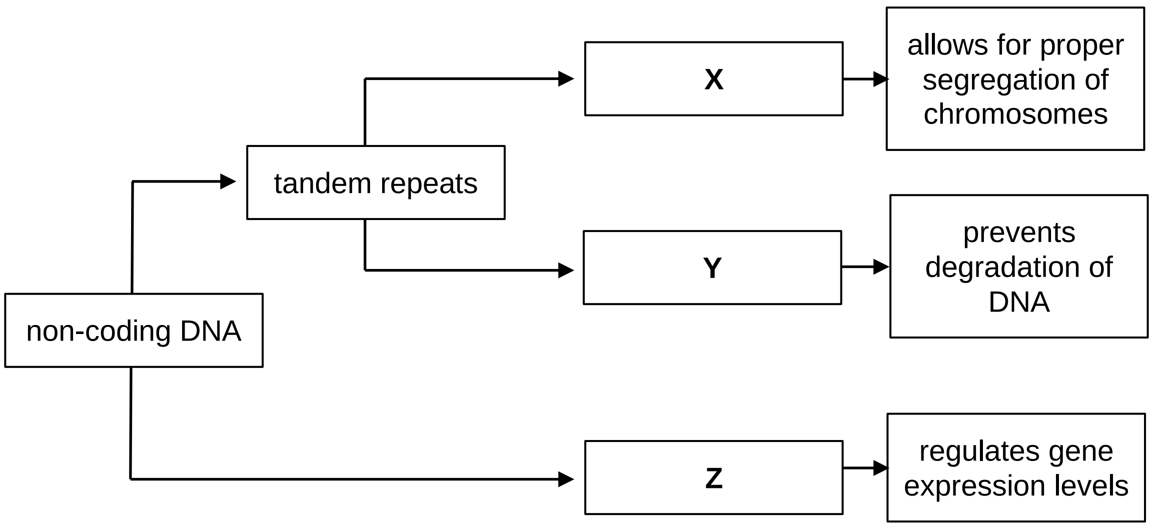
D Inhibit the attachment and thereby entry of the virus into its host cell by developing antibodies that bind to the haemagglutinin on the viruses.

SC: Influenza viral reproductive cycle
OR:
- ✗ A.Translation to produce normal host cell proteins will also be affected.
- ✗ B: Transcription to produce normal host cell RNA will also be affected.
- ✗ C: By binding to host cell sialic acid is not an option as sialic acid is ubiquitous and blocking all sialic acid is not practical.
- ✓ D: Yes, by targeting HA, virus cannot adsorb on host cell and thus cannot enter host cell.

Ans: D

[L3] (novel)

14 The flowchart shows the classification of several regions of non-coding eukaryotic DNA, **X**, **Y** and **Z**.



Which statement(s) correctly describes **X**, **Y** and **Z**?

1 Regions **X** and **Y** are made up of transcriptionally active tandem repeats.

2 Regions **X** and **Y** are always associated with proteins, but DNA at region **Z** is only associated with proteins during gene expression.

3 Region **Z** may involve DNA bending but region **Y** shortens during DNA replication.

4 Regions **X**, **Y** and **Z** are conserved throughout the life of the organism.

A 2 only

B 3 only

C 1 and 4 only

D 2 and 3 only

SC: X - centromeres, Y – telomeres and Z – control elements (non-coding)
OR: Statement 1 False (all non-coding), Statement 2 False (all associated with protein, Z is associated with proteins during chromatin packaging), Statement 3 True, Statement 4 False (not all conserved).

Ans: **B**

[L3] (HCI 2015 Prelim P1 Q14 modified)

15 Which one of the following statements correctly describes the role of enhancers.

A DNA sequences that are bound by general transcription factors.

B DNA sequences that directly induces the bending of DNA.

C DNA sequences that are involved in stabilisation of the transcriptional initiation complex.

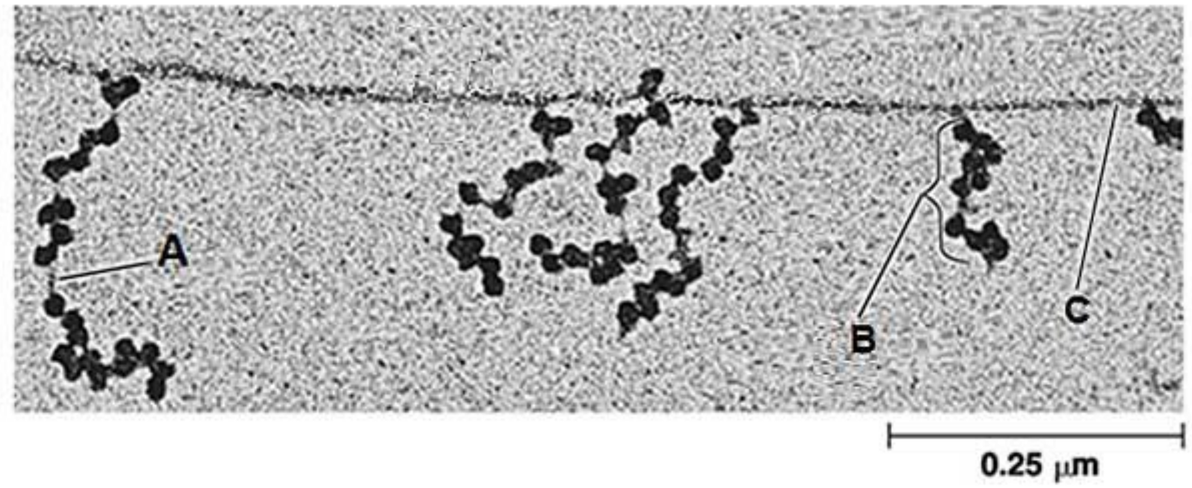
D DNA sequences are proximal control elements that are non-coding.

SC: Factual recall
OR:
- A is wrong because it should not be general transcription factors.
- B is wrong because it does not directly induce bending but bending occurs only after an activator binds to it.
- C is correct.
- D is wrong because it is not a proximal control element.

Ans: **C**

[L1]

16 The electron micrograph below shows several labelled structures present in a mitochondrion.



Which of the statements below correctly describe the labelled structures?

1 The structure labelled **A** is the polypeptide chain.

2 The structures labelled **B** are polyribosomes which consist of many 70S ribosomes.

3 The structure labelled **C** is the 3' end of template DNA strand.

4 The structure labelled **C** is the 5' end of the mRNA strand.

A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

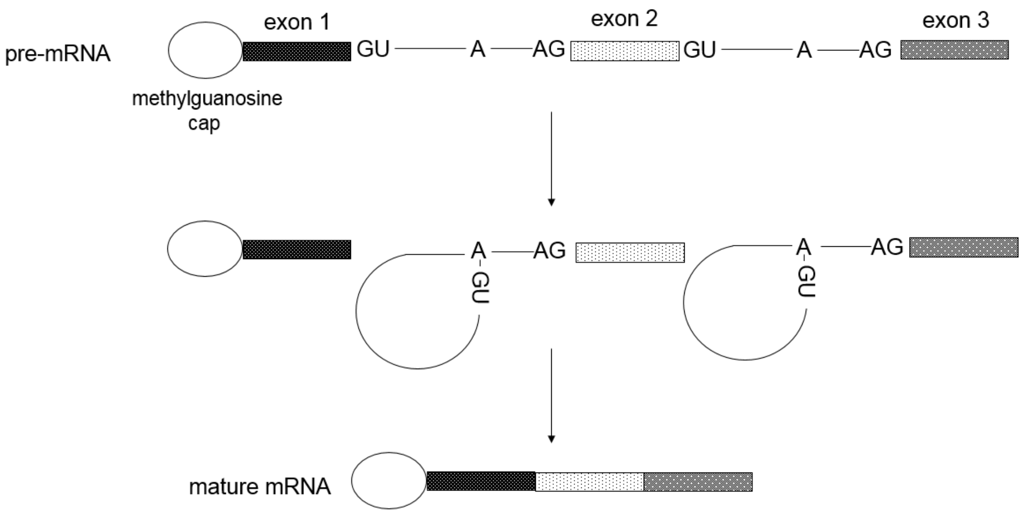
D 1, 2 and 4 only

SC: Labelled structure in EM is polyribosomes (structure B), ribosomes binds to mRNA template for translation (hence structure A is mRNA), Structure C must be DNA template strand. 70S ribosome because context is mitochondrion.
OR: Statement 1 false; Statement 2 true; Statement 3 true; Statement 4 false → accept Option C

Ans: **C**

[L3] AJC 2015 Prelim P1 Q16

17 The diagram shows part of an mRNA undergoing the process of splicing.



With reference to the diagram above, which statement(s) is / are related to the process shown?

1 RNA splicing occurs after the release of pre-mRNA from RNA polymerase.

2 Spliceosome binds to the 3' splice site GU and the 5' splice site AG on the pre-mRNA.

3 A RNA loop is formed on the pre-mRNA where the intron is excised.

4 There can be more than one type of product formed from a single pre-mRNA.

A 1 and 2

B 3 and 4

C 2, 3 and 4

D 1, 3 and 4

SC: Diagram shows RNA looping, splicing (alternative splicing possible)
OR: Statement 1, 3 and 4 possible (even if not shown explicitly as qns ask for related process) Statement 2 is wrong (should be 5'GU and 3'AG)

Ans: D

[L3] (HCI 2015 Prelim P1 Q15)

18 Gene expression is similar in prokaryotes and eukaryotes in that both:

A have post-transcriptional modifications.

B require helicase to separate the DNA so that transcription can take place.

C have spliceosomes to intron splicing.

D involve attachment of proteins to DNA adjacent to the gene being transcribed.

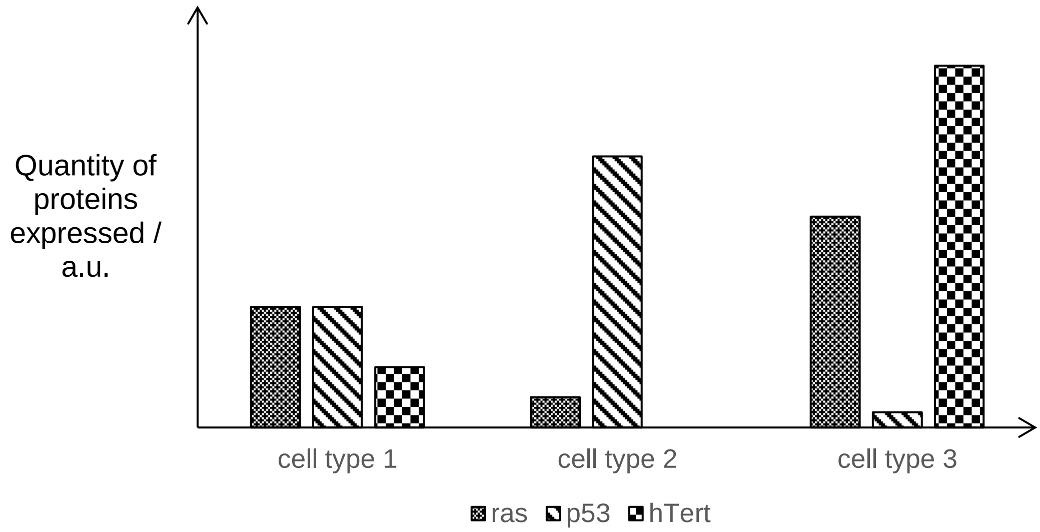
SC: Transcription and translation are common to both prokaryotes and eukaryotes. Among options given, points on transcription are given.
OR: Statement A false (prokaryotes do not have), Statement B false (RNA polymerase not helicase), Statement C (Prokaryotes do not utilise spliceosome). Accept D.

Ans: **D**

[L2]

19 Cancer critical genes include ras, p53 and hTert. hTert codes for human telomerase.

The levels of proteins expressed by each gene in three different cell types of a patient are shown in the graph. Only one cell type was taken from a malignant tumour.



Which statement is true?

A Cell type 1 is not from the malignant tumour since balanced expression of *ras* and *p53* halts cell cycle progression.

B Activation of telomerase will result in cell type 2 gaining immortality and becoming cancerous.

C Cell type 3 is obtained from the malignant tumour as the cells will divide uncontrollably.

D Gain-of-function mutation of *hTert* in cell type 1 will result in malignant tumour formation.

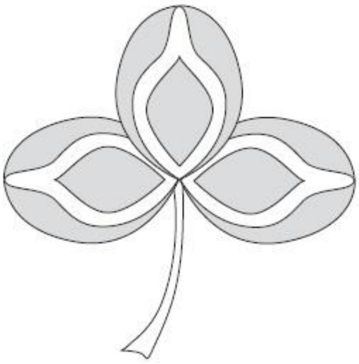
SC: ras – oncogene, p53 – Tumour suppressor gene, hTert – Telomerase gene (activated in cancer)

OR: Statement A false (insufficient info from data), Statement B false (no expression of telomerase), Statement D false (insufficient info from data to conclude gain-of-function mutation). State C true as it fulfils 3 conditions for cancer development (overexpression of ras oncogene, underexpression of p53 tumour suppressor gene and high expression of telomerase).

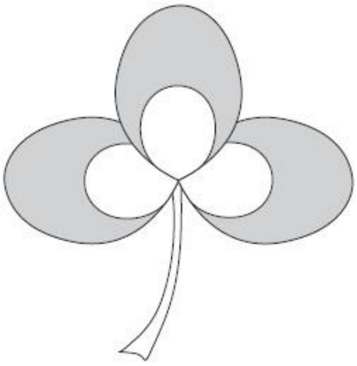
Ans: **C**

[L2] (HCI 2015 Prelim P1 Q17 modified)

20 The white clover, *Trifolium repens*, is one of the plants found growing as a weed in many lawns. Leaves of the white clover are divided into three leaflets, which often have characteristic white patterns visible on their surface. The two basic forms of the pattern are a chevron and patch. The diagram below shows these two patterns.

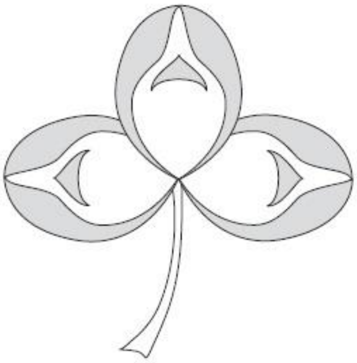


chevron pattern



patch pattern

If a pure-breeding clover plant with the chevron pattern is crossed with a pure-breeding plant with the patch pattern, the offspring have leaflets with a mixed chevron and patch pattern, as shown in the diagram below.



mixed pattern

Which row correctly describes the inheritance of leaflet patterns in white clover?

	number of alleles that determines the white patterns in the leaflets	mode of inheritance
A	2	codominance
B	2	epistasis
C	> 2	codominance
D	> 2	epistasis

SC: only 2 alleles (one for chevron and other for patch). Offspring shows mixed pattern (both expressed hence codominace)

OR: Accept A, Reject B (not epistasis), C (not sufficient information provided to infer 3 or more alleles) and D (not epistasis)

Ans: **A**

[L2] (HCI 2015 Prelim P1 Q19)

21 In a cross involving polygenic inheritance, 3 genes control the height of a tulip plant. The shortest and tallest plants are 12 cm and 24 cm respectively.

Assuming all other environmental factors are kept constant, what is the height of the F1 offspring obtained from a cross between a homozygous 12 cm and a homozygous 24 cm plant?

A 6 cm

B 12 cm

C 14 cm

D 18 cm

SC: polgenic inheritance involving 3 gene A,B,C. shortest homozygous (aabbcc – 12cm → every recessive allele contribute 2 cm) tallest homozygous (AABBCC - 24cm → every dominant allele contribute 4 cm)

OR: Cross between homozygous recessive (aabbcc) and homozygous dominant (AABBCC) result in heterozygous (AaBbCc) which will result in 18 cm tall plants.

Ans: D

[L3] (VJC 2015 Prelim P1 Q20 modified)

22 A plant with orange-spotted flowers was grown in a greenhouse from a seed collected in the wild. The plant was self-pollinated and gave rise to the following progeny: 129 plants with orange-spotted flowers, 22 plants with yellow-spotted flowers, 26 plants with solid orange flowers, and 15 plants with solid yellow flowers.

The formula for the chi-squared (χ^2) test is given as follows:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

degrees of freedom	probability			
	0.10	0.05	0.01	0.001
1	2.71	3.84	6.64	10.83
2	4.69	5.99	9.21	13.82
3	6.25	7.82	11.35	16.27
4	7.78	9.49	13.28	18.47

Which statement is true about the inheritance of flower colour and flower pattern at 99% confidence level?

A Since $p < 0.05$, the difference between the observed and expected results is not significant. The inheritance of flower colour and flower pattern is following Mendel's law of independent assortment.

B Since $p > 0.05$, the difference between the observed and expected results is not significant. The inheritance of flower colour and flower pattern is not following Mendel's law of independent assortment.

C Since $p > 0.01$, the difference between the observed and expected results is not significant. The inheritance of flower colour and flower pattern is following Mendel's law of independent assortment.

D Since $p < 0.01$, the difference between the observed and expected results is significant. The inheritance of flower colour and flower pattern is not following Mendel's law of independent assortment. .

SC: Observed no. compared against dihybrid ratio of 9:3:3:1, chi square calculated for df=3 is 6.91.

OR: p>0.05 (random chance of difference high), not statistically significant, inheritance follows Mendel's law of independent assortment.

Ans: C

[L2] (HCI 2015 Prelim P1 Q 23 modified)

23 Which of the following statements about transport in the cell is incorrect?

A Active transport is the movement of substances across the cell membrane against a concentration gradient.
B Diffusion is the mechanism by which movement of hydrophobic particles through a cell membrane down a concentration gradient.
C Receptor mediated endocytosis involves the binding of the substance to specific receptors and their subsequent passive entry into the cell.
D Bulk transport is a process which requires energy.

Hydrophobic
Hormone
Steroids

BARRIER

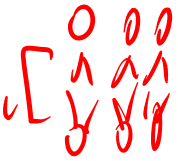
SC: transport / false / transport / correct
OR: A True

B True

C False (Receptor mediated endocytosis is an active process.)

D True

H_2O phobic



Ans: **C**

[L1]

24 Which of the following statements is false about cell signalling involving tyrosine kinase receptors.

A Ligand molecules are mostly hydrophilic in nature.

B Different activated relay proteins serve to directly amplify the effects of the ligand.

C Dimerisation serves to initiate auto-phosphorylation.

D Receptors are transmembrane proteins that are anchored within the cell surface membrane.

SC: statement / false / TKR
OR: A True (are extracellular within an aqueous medium) True

B Different activated relay protein affect different processes and therefore are not seen to directly amplify
(ie there is more of the same effect) False
C Correct True
D both hydrophobic tails and hydrophilic head form anchor points with the respective portions of the receptor True

Ans: **B**

[L2] Novel

25 Phosphorylation cascade is an important component in cell signaling, which of the following statements is incorrect about this cell signaling mechanism.

A The signal is passed via a series of phosphorylation involving protein kinases.

B The mechanism allows for greater control and speed in transmission of the signal.

C Phosphorylation cascade mechanism is initiated by a second messenger.

D Inactivation of the signal mechanism involves phosphatases which deactivate protein kinases.

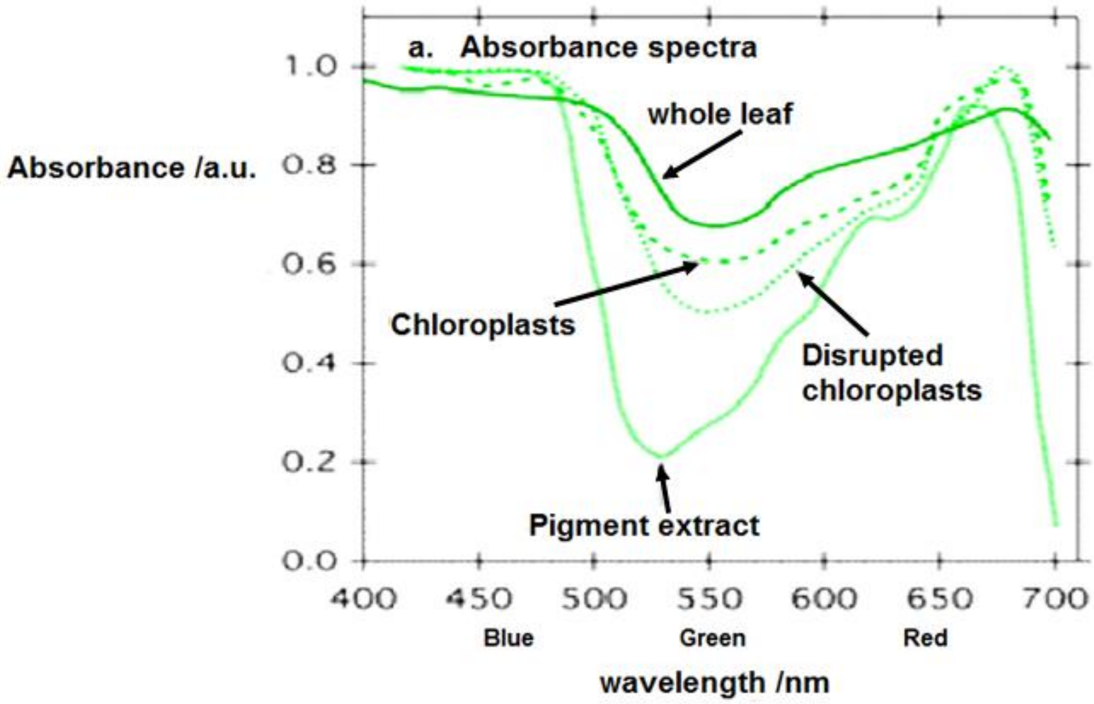
SC: statement / false / phosphorylation cascade
OR: A True
B False -Phosphorylation cascade does not speed up the transmission of the signal.
C True
D True

Ans: **B**

[L1]

21

26 The figure shows the absorptance spectra of various components of a leaf.



What can be inferred from the data shown?

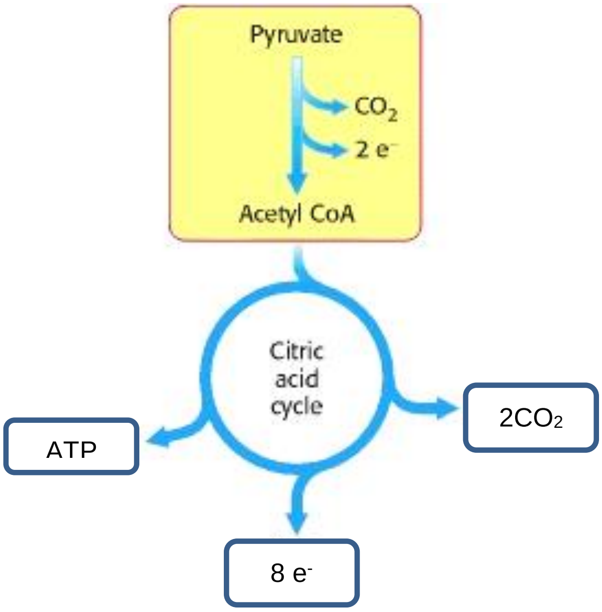
A Absorptance is highest in 650-700 nm in all leaf components due to light harvesting complexes.
B Whole leaf samples experience the least absorptance at 550 nm due to all green light being reflected.
C Disrupted chloroplast samples have higher absorptance compared to chloroplasts due to a larger surface area for light capture.
D Pigment extracts are the main agents of light harvesting due to the presence of carotenoids.

SC: inferred from data
OR: A × except in whole leaf where at 475nm, absorption is higher compared to 650-700nm.
B × not all green light is reflected, there is still absorption of 0.2 a.u.
C × Disrupted chloroplasts lack the organization compared to chlorophyll.
D ✓ Carotenoids make up the main light harvesting components of the leaf.

Ans: **D**

[L3]

27 The figure below shows the part of the process of aerobic respiration.



Which of the following statements is true of the significance of acetyl CoA?

A Acetyl CoA is the product of the link reaction and is subsequently brought into the mitochondria to enter the Krebs cycle

B Acetyl CoA is the entry point into the metabolic pathway of both carbohydrates and fats.

C Acetyl CoA is an energised molecule combined with Oxaloacetate, to yield 4 molecules of citric acid per molecule of Glucose.

D Electrons released in the formation of Acetyl CoA are used in the production of NADPH.

SC: true significance / acetyl CoA

OR: A × link reaction occurs in the mitochondrial matrix.

B ✓ lipid metabolism enters here.

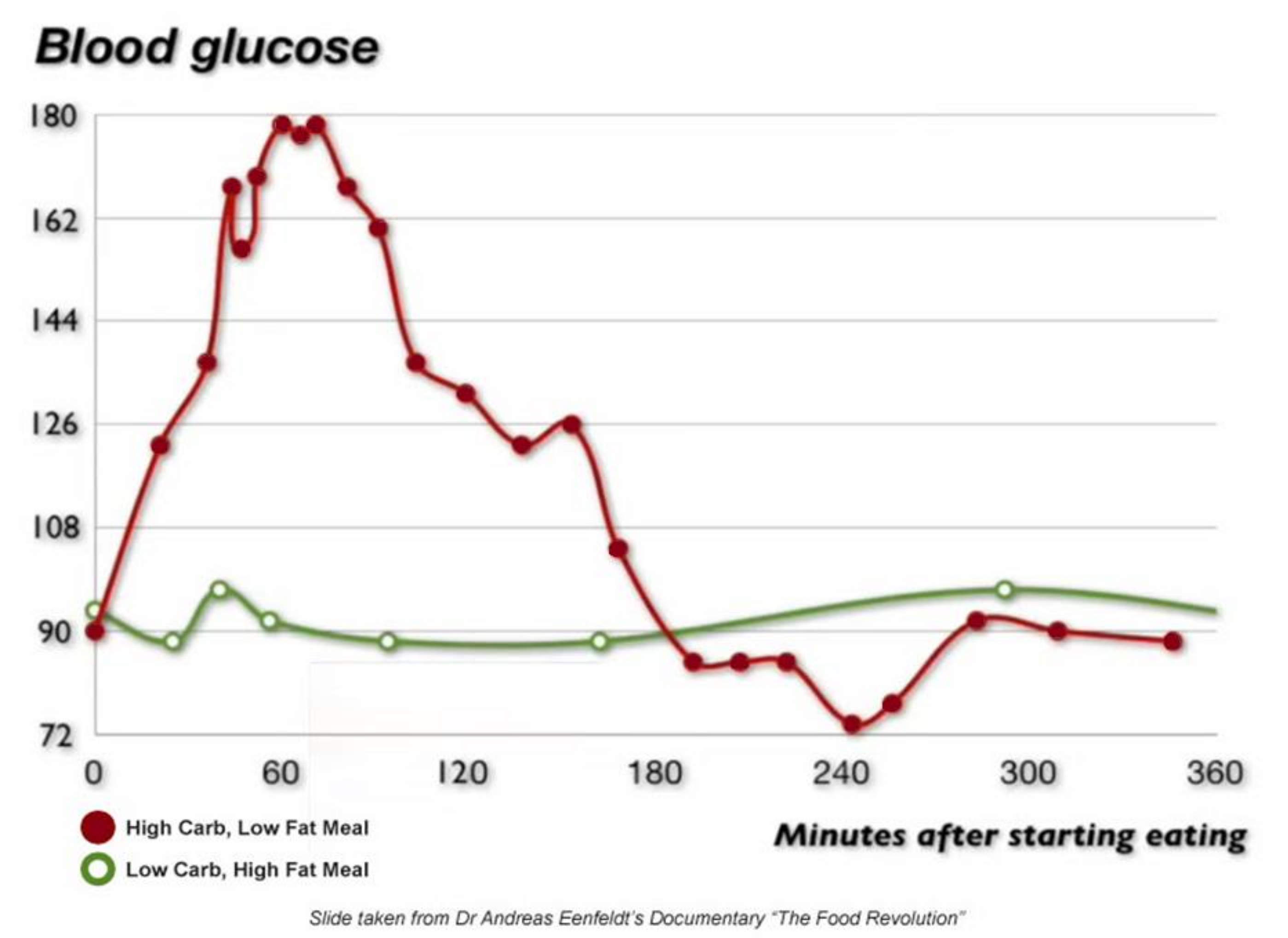
C × only 2 molecules of citric acid are produced for every 1 molecule of glucose.

D × electrons released is used in the production of NADH not NADPH.

Ans: **B**

[L2]

28 The figure below represents the blood glucose levels of a normal person after a meal.



At which time do the beta cells of the islets of Langerhans detect and effect the secretion of insulin to manage blood glucose levels?

	Detection by beta cells	Secretion of insulin
A	0 to 5 minutes	61 to 180 minutes
B	0 to 5 minutes	0 to 10 minutes
C	180 to 200 minutes	240 to 300 minutes
D	180 to 200 minutes	180 to 200 minutes

SC: time / beta cells Islets Langerhans / detect and secrete insulin

OR: A × detection and secretion are close together –detection immediately followed by secretion of insulin

B ✓ as explained in A.

C × detection is by Alpha cells of islets of Langerhans and for hypoglycaemic conditions..

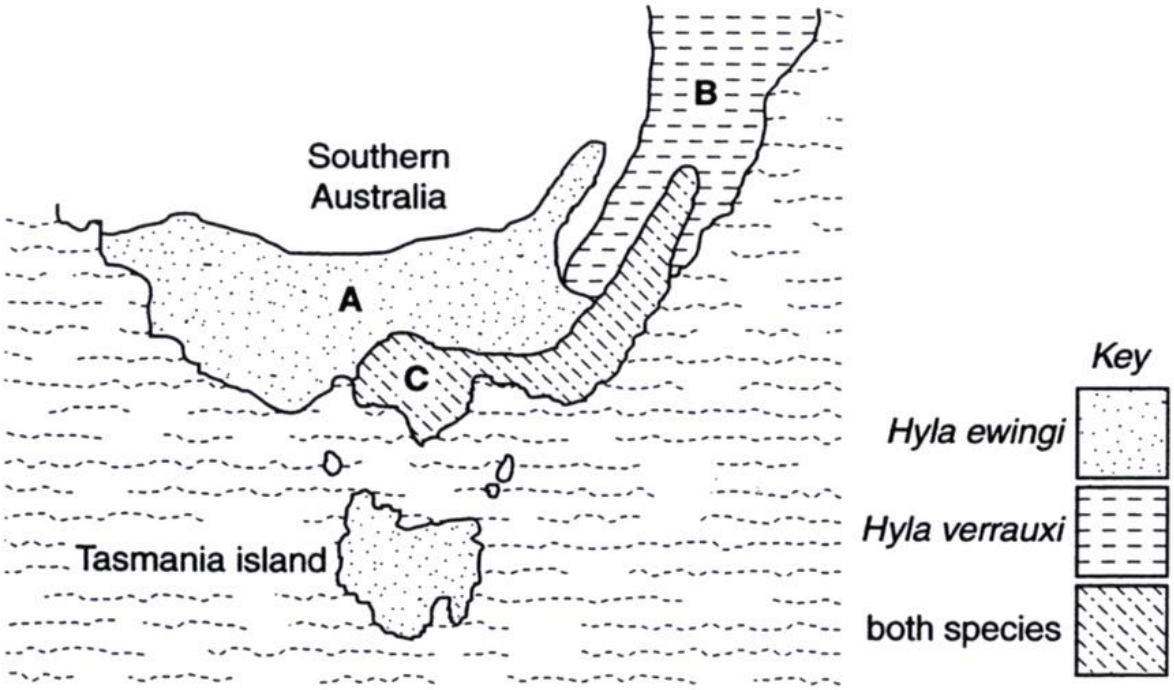
D × detection and secretion are close together- detection immediately followed by secretion of glucagon.

Ans: **B**

[L1 high/ L2]

23

29 Two closely related species of frog, *Hyla ewingi* and *Hyla verrauxi* live in South Australia. The figure below shows the distribution of the tree frogs in Southern Australia.



Hyla ewingi and *Hyla verrauxi* are two closely related species of tree frogs from southern Australia. Research from breeding studies and DNA sequence data has shown that they have weak genetic incompatibility.

Male frogs attract females of the same species for mating by their pulsing call. The pulse rate of the male calls of the two species is almost identical. However, when both species coexist within the same region, the calls of *H. ewingi* are quite different than those of *H. verrauxi*.

Which of the following can be correctly inferred from the data given?

A Complete speciation has taken place between the two groups of frogs.

B Allopatric speciation was probably the evolutionary mechanism at work.

C Convergent evolution has seen the frogs in Tasmania similar to those in region A in Australia.

D Sympatric speciation was probably the evolutionary mechanism at work.

SC:
1. P2L1 weak genetic incompatibility – ie genetically similar
2. P3L1 Male attract females for mating – ie are not reproductively isolated- same species
3. P3L1 different behaviours- behavioural isolation beginning but not led yet to reproductive isolation.
4. P3L2 coexist in same region, no geographical isolation- Sympatric speciation occurring over time.

-correctly inferred / data given

OR: A × reasons from point 2.

B × reasons from point 4.

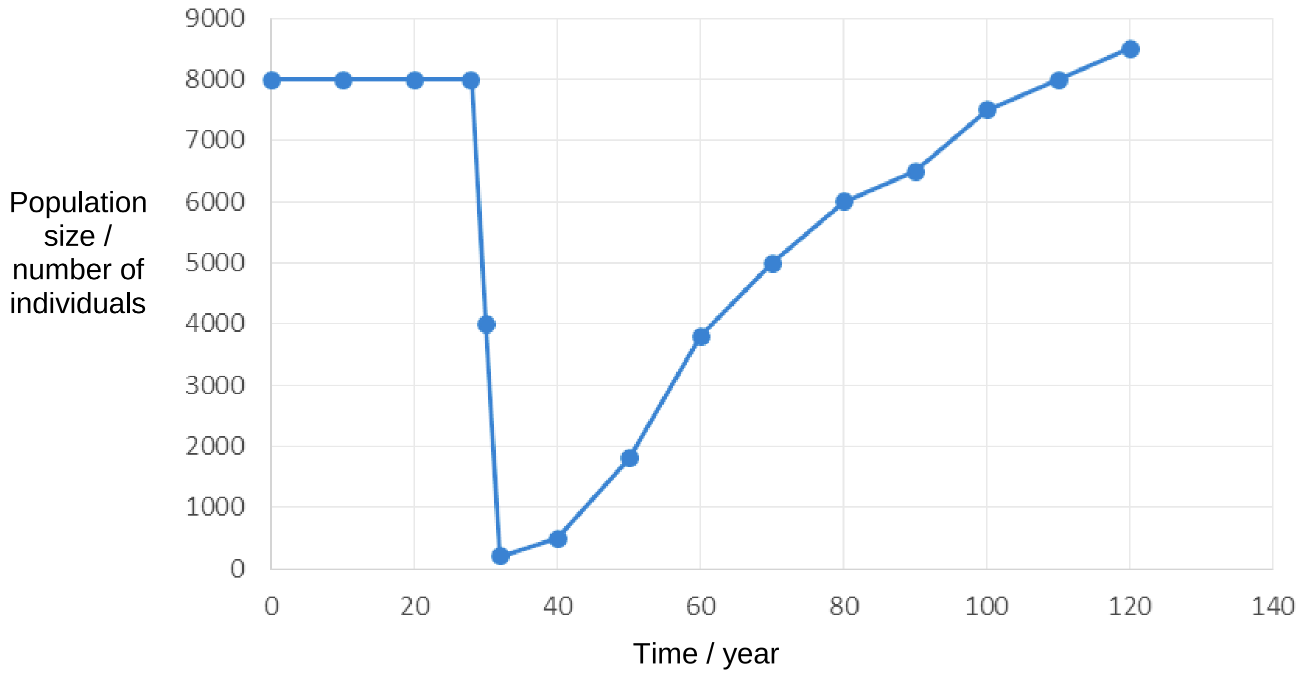
C × plate tectonics in biogeography would explain H. ewingi present in both regions not convergent evolution. Besides they are exactly the same species so cannot be Convergent evolution.

D ✓reasons from point 4.

Ans: **D**

[L2]

30 The figure shows the population of a group of organisms in a fixed region over time.



The following statements are derived from the data shown.

i. Bottleneck event has taken place Year 30 to 35
ii. Genetic variation has been fully restored by Year 110
iii. Allele frequency steadily increases due to genetic drift
iv. Founder effect has taken place from Year 28 onward

Which statements can be concluded as true?

A i only

B ii only

C i, ii & iii only

D i, iii & iv only

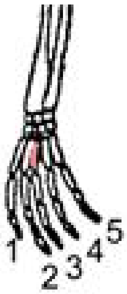
SC: 1. P1L1 Populatio .. fixed region ie no founder effect
OR: 2. i ✓ Bottle neck possible drastic reduction in population
3. ii × genetic variation is reduced after bottleneck and is only increased by mutation over time but would not have been regained by year 150.
4. iii × Not conclusive from the table.
5. iv × Founder Effect is not considered in the original population see point 1.

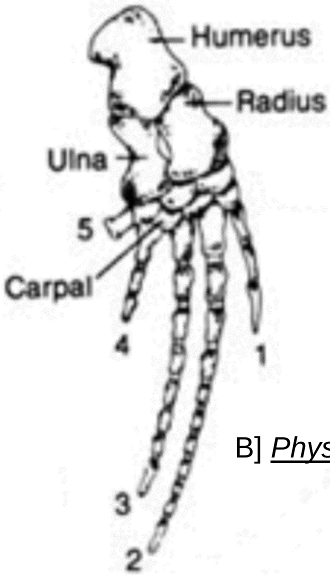
Ans: **A**

[L2 high]

31 The following figure shows the anatomy of the left front appendages of the two vertebrates.

A] <u>*Javan surili*</u>





B] <u>*Physeter macrocephalus*</u>

Which one of the following correctly describes the type of structures seen and their evolutionary connection?

	Type of structures	Ancestry	Type of evolution
A	Homologous	Different ancestor	Convergent Evolution
B	Homologous	Common ancestor	Divergent Evolution
C	Analogous	Different ancestor	Convergent Evolution
D	Analogous	Common ancestor	Divergent Evolution

SC: correct type structure / evo connection
OR: Homologous (same pentadactyle plan) / divergent evo / common ancester
A × same ancestor / not convergent
B ✓ correct.
C × not analogous / same ancestor / not convergent.
D × not analogous / not divergent evo.
Ans: **B**

[L2]

32 All the characteristics below support the use of plasmids in cloning and expression in bacteria **except**

A More than one plasmid can be taken up by each bacterium.

B Plasmids are able to control their own replication through their origin of replication

C Plasmids contain a wide range of restriction sites for various restriction enzymes.

D Linker DNA / artificial sticky ends are not required to express eukaryotic genes of interest derived from cDNA. .

SC: factual recall
OR: Statement A, B and C are true. Only Statement D false. Accept option D.

Ans: **D**

[L1] (VJC 2015 Prelim P1 Q33 modified)

33 *lacZ* gene is a genetic marker found in the plasmid which can be used in genetic engineering.

What is the function of *lacZ* gene in a cloning vector?

A Express *lac* repressor

B Distinguish between introns and exons

C Break down lactose to galactose and glucose

D Screen for cells with the recombinant plasmid .

SC: factual recall
OR: Statement A wrong, B (wrong) and C (true but not align to qn context. Also, the gene does not do the breakdown of lactose, it is the enzyme that is encoded for). Only Statement D relevant. Accept option D.

Ans: **D**

[L1] (TJC 2015 Prelim P1 Q33)

34 Which correctly describe the difference between genomic and cDNA libraries and the corresponding reason for the difference?

	Difference	**Reason**
A	Genomic library may contain distant control elements and centromere while cDNA library contains only expressed genes.	Due to mRNA splicing which removes exons and splice introns together.
B	Genomic library is larger in size than cDNA library.	Due to genomic library containing total DNA extract from a cell while cDNA library containing the total mRNA extract from a particular cell type.
C	Genomic library from an individual is always the same, while cDNA library may differ, depending on the cell type from which it is constructed.	Due to differential gene expression in the different cell types.
D	Genes from genomic library are more suited than those from cDNA library for expression in bacterial cells.	Due to the action of restriction endonucleases in the construction of genomic library and that of reverse transcriptase in the construction of cDNA library.

SC: Statement Evaluation of DNA libraries

OR: Statement A wrong (introns are excised and exons spliced), B (cDNA library does not contain total mRNA extract) and D (genes from genomic library are not more suitable for expression in bacterial cells due to presence of introns). Only Statement C True. Accept option C.

Ans: **C**

[L2] (MJC 2015 Prelim P1 Q37)

35 Genes P, Q, R and S occur on the same chromosome. The table shows the recombination frequencies.

	recombination frequency (%)
between P and Q	46
between P and R	8
between R and Q	54
between Q and S	13
between R and S	41

Which of the following represents the correct order of genes on the chromosome?

A P – Q – R – S

B P – R – S – Q

C R – P – S – Q

D R – S – Q – P .

SC: Distance between gene loci can be inferred from recombination frequencies; the closer the gene loci → smaller recombination frequency, the more distant gene loci → higher recombination frequency.
OR: R and Q has the highest recombination frequency so should be furthest from each other → Option C

Ans: **C**

[L2] (IJC 2015 Prelim P1 Q35)

36 The Human Genome Project facilitated genetic testing of individuals and renewed emphasis on ethical and social implications.

Which of the following statements correctly describe unintended consequences of genetic testing?

1 discovery of wrongly attributed paternity

2 unauthorised publication of genetic test results

3 psychological stress after receiving genetic test results

4 social stigmatisation of genetically predisposed individuals

A 1 and 2

B 3 and 4

C 1, 2 and 3

D All of the above .

SC: For HGP genetic testing, all are unintended.
OR: Statement 1, 2, 3 and 4 are unintended consequences. Accept Option C.

Ans: **D**

[Low L2]

37 Which of the following shows the correct developmental potency of the following stem cells.

	Haematopoietic stem cells	Zygotic stem cells	Embryonic stem cells	Neural stem cells
A	Multipotent	Pluripotent	Totipotent	Unipotent
B	Multipotent	Totipotent	Pluripotent	Unipotent
C	Multipotent	Pluripotent	Totipotent	Multipotent
D	Multipotent	Totipotent	Pluripotent	Multipotent

.

SC: Stem cells' developmental potency
OR:
- HSCs & NSCs are multipotent
- Zygotic stem cells are totipotent
- ESCs are pluripotent

Ans: D

[L1] (novel)

38 Which problem is associated with gene therapy?

A Target cells do not have suitable receptors on their cell surface membranes.

B The nuclear pores do not allow the vector into the nucleus.

C The viral vector cannot trigger cyclic AMP to activate appropriate genes.

D Viral vectors insert therapeutic genes at random points in the genome. .

SC: Stem cells' developmental potency
OR:
- A: Receptors are not necessarily required. Non-viral methods do not require receptors.
- B: They usually do.
- C: cAMP not necessarily triggered.
- D: Yes, this will cause insertional mutagenesis and hence cancer.

Ans: D

[L2] ('13 A-level/P1/Q39)

39 Some plant tissue culture techniques involve the production of protoplasts using leaf tissue. Preparation of protoplasts involves incubation in a solution that contains enzymes such as cellulase and pectinase.

Which statement about protoplast is not correct?

A Protoplasts are pluripotent and regenerate into whole plants when provided with the correct growth factors.

B Protoplasts are more susceptible to microbial contamination.

C Protoplasts are used for the production of genetically engineered plants as they take up naked DNA easily.

D Protoplasts need to be maintained in the solution of the same water potential to prevent lysis. .

SC: Plant cloning
OR:
- A: They are not pluripotent. They can be totipotent.
- B: Yes they are.
- C: Yes they take up naked DNA easily as they lack cell wall.
- D: Yes.

Ans: A

[L2] ('15 HCI/P1/Q40)

40 Which statement supports the view that genetically engineered animals could help to solve the demand for food in the world?

A Transgenic pigs and sheep are produced to express higher levels of growth hormone.

B Biomedical applications of genetically engineered animals have also become routine within the pharmaceutical industry, for drug discovery, drug development and risk assessment.

C Cloning of either extinct or endangered species such as thylacine and woolly mammoth helps to retain genetic diversity in small populations.

D By inserting genes from sea anemone and jellyfish, zebrafish have been genetically engineered to express fluorescent proteins.

SC: GMO animals
OR:
- A: Correct.
- B: Does not solve the demand for food.
- C: Does not solve the demand for food.
- D: Does not solve the demand for food.

Ans: A

[L2] ('15 HCI/P1/Q40)

END OF PAPER